Spring 2025-26

Operating System(CSC-204)

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Time Table

- Lectures:
  - 10.00 – 10.55 hours IST (Tuesday) (GB 205)
  - 10.00 – 10.55 hours IST (Thursday) (GB 205)
  - 10.00 – 10.55 hours IST (Friday) (GB 205)

- Tutorials / Quizzes / Discussions / Doubts / etc.
  - 14.00 – 14.55 hours IST (Wednesday) (GB 105)
  - 15.00 – 15.55 hours IST (Wednesday) (GB 105)

- Ad-hoc Teaching Assistants (TAs)
  - Swati Singh
  - Shikha Arya
  - Parth Srivastava

Course Overview

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

NAME OF DEPT./CENTRE: **Computer Science and Engineering**

1. **Subject Code: CSC-204** Course Title: **Operating Systems**

2. Contact Hours: **L: 3** **T: 1** **P: 0**

3. Examination Duration (Hrs.): **Theory: 3** **Practical: 0**

4. Relative Weightage: **CWS: ~~2[illegible]0~~** **PRS: 0** **MTE: ~~2[illegible]0~~** **ETE: ~~4[illegible]0~~** **PRE: 0**

5. Credits: **4** 6. Semester: **Spring** 7. Subject Area: **DCC**

8. Pre-requisite: **Computer Architecture and Microprocessors**

9. Objective: To provide an understanding of the functions and modules of an operating system and study the concepts underlying its design and implementation.

Weightage

1. Subject Code: **CSC-204** Course Title: **Operating Systems**

2. Contact Hours: **L: 3** **T: 1** **P: 0**

3. Examination Duration (Hrs.): **Theory** 0 3 **Practical** 0 0

4. Relative Weight: **CWS** **PRS** 00 **MTE** **ETE** **PRE** 00

5. Credits: 0 4

6. Semester **Autumn** ✓ **Spring** **Both**

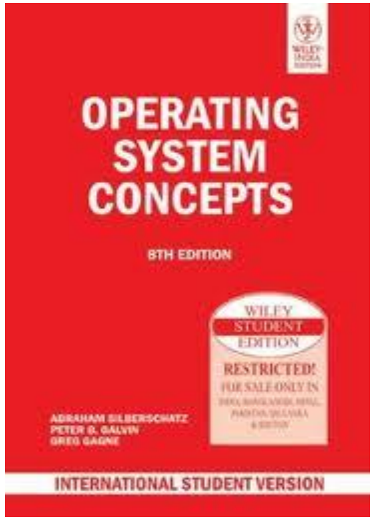
- CWS – 30%
  - Attendance: 30-40%
  - Quizzes: 40-50%
  - Assignments: 20-30%
- MTE – 30%
- ETE – 40%

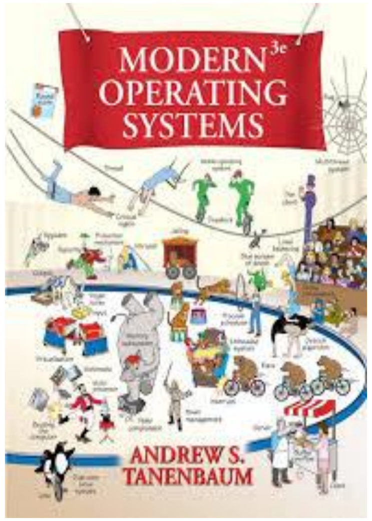
Thematic View of the Syllabus

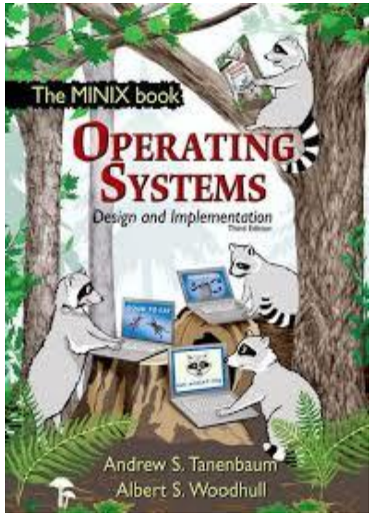
Sl. No.	Contents
1.	**Fundamental Concepts of Operating System:** Operating system functions and characteristics, historical evolution of operating systems, issues in operating system design. Basic Overview of Windows, Unix, and Linux
2.	**Process Management:** Process abstraction, process address space, process management, system calls, threads, process hierarchy.
3.	**CPU Scheduling:** Levels of scheduling, comparative study of scheduling algorithms, multiple processor scheduling.
4.	**Deadlocks:** Characterization, prevention and avoidance, deadlock detection and recovery.
5.	**Concurrent Processes:** Critical section problem, semaphores, monitors, inter-process communication, message passing mechanisms.
6.	**Memory Management:** Storage allocation methods, virtualization and virtual memory concepts, demand paging, page replacement algorithms, segmentation, thrashing
7.	**File Systems:** Functions, file access and allocation methods, directory system, file protection mechanisms, implementation issues, file system hierarchy
8.	**Device Management:** Hardware organization, device scheduling policies, device drivers.
9.	**Security:** Threats and Attacks, Viruses: Memory Resident Viruses, Boot Sector Viruses, Root kits, Authentication, Access Control Lists, Capabilities, Anti-Virus & its techniques.

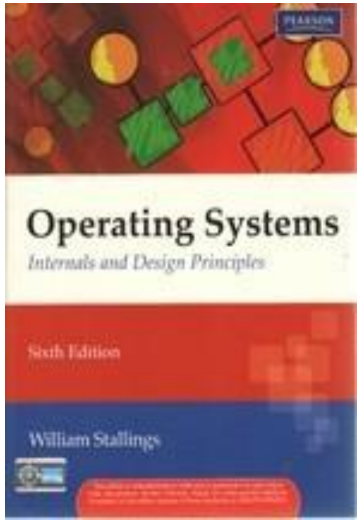
Books

- Operating System concepts
  - Silberscharz and Galvin   (Publishers: Wiley)
- Modern Operating Systems
  - Tanenbaum (Publisher: Pearson)
- Operating Systems
  - William Stallings (Publishers: Pearson)
- Operating Systems
  - Tanenbaum (Publishers: Pearson)
- Distributed Operating Systems
  - Tanenbaum(Publishers: Pearson)
- Operating Systems
  - Gary Nutt (Publishers: Pearson)
- System Programming and Operating Systems
  - D.M. Dhamdhere (Publisher: TMH)







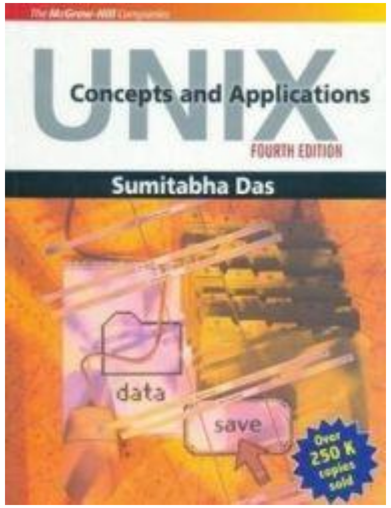


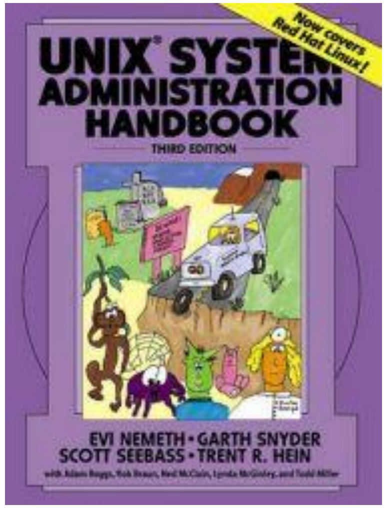
Books

- The UNIX Programming Environment
  - Brian Kernighan (Publishers: PHI)
- Understanding LINUX kernel
  - Daniel P Bovet, Macro Cesati (Publishers : O'Reilly)
- Unix Concepts and Applications
  - Sumitabha Das (Publisher: TMH)
- Unix Shell Programming
  - Kanetkar (Publisher: BPB)
- Unix System Administration Handbook
  - Evi Nemeth (Publisher: Pearson)

Labs

Beginners & Laboratory





Important Instructions

- Negative marks for Plagiarism/Copying. **I mean it!!**
- All deadlines for submissions are hard. Helpless even if you have a genuine reason.
- Absence to the Quizzes, missing deadlines, or missing or violating any such course requirements will be treated seriously. No marks will be awarded on any case. No exceptions in any form.
- Dress code: Formals in class.
- No Calling/Messaging/emailing directly to me. I DO NOT RESPOND. Only MS Teams or CR.
- **Caution:** Class size is large. You may miss the grade by a half mark.
- Default: Closed Book Examinations !!
- Expect Surprise Quizzes.

Welcome Again!

- Have a nice day!!
- Happy Semester !!!

Class 0:

- Learn to install at least one
  - Windows version of the operating system
  - Linux version of the operating system
    - Preferably RHL
- Preferably, on dual boot (**Word of Caution!**)

- Learn/Practice basic UNIX / LINUX commands